

Avranil Dhar

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SUMMARY

- Engineering student applying Python, JavaScript, and Java to architect and develop full-stack web applications and developer tooling. Builds interactive, responsive frontends with React and implements backend services using Node.js and REST APIs. Focuses on optimizing developer experience, reducing configuration overhead, and delivering efficient, user-centric solutions while maintaining a strong academic record.

TECHNICAL SKILLS

- ▷ **Languages:** C, Java, JavaScript, Python
- ▷ **Frontend:** HTML5, CSS3, Tailwind CSS, React.js
- ▷ **Backend:** Node.js, Bun, Express.js, REST API Development, LangChain
- ▷ **Databases & Tools:** MySQL, MongoDB
- ▷ **Tools & Platforms:** Git, GitHub, Linux, Postman, Clerk, Vercel
- ▷ **Computer Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming, Computer Networks, Operating Systems

PROJECTS

ContextAI– Retrieval-Augmented Generation Document Intelligence System [GitHub](#) [Live](#) Jun 2026 - Jul 2026

- Engineered an enterprise-grade RAG pipeline using **TypeScript**, **LangChain**, **Google Gemini**, and **Groq** with **MongoDB Vector Search** for high-accuracy document retrieval and context-aware LLM responses.
- Architected an asynchronous multi-format document ingestion engine using **BullMQ**, **Redis**, **unpdf**, and **mammoth** with **RecursiveCharacterTextSplitter** for seamless PDF/DOCX parsing and optimal embedding chunking.
- Built a full-stack, type-safe web application featuring **React 19**, **Vite**, **Tailwind CSS v4**, and **Express.js** integrated with **Better Auth** for secure session management and access control.

Ionospheric TEC Forecasting & Ionospheric Delay Correction (Group Project) [GitHub](#) Mar 2026 – Present

- Leading a mentored group project forecasting Total Electron Content (**TEC**) 24 hours ahead to mitigate ionospheric propagation delays in satellite navigation (**GNSS**), utilizing real **NavIC/IRNSS** satellite datasets from IISc Bangalore.
- Architected and benchmarked stacked **LSTM** and **Transformer** seq2seq models in **PyTorch** for day-ahead 1440-step TEC forecasting; derived L1 and L5 band ionospheric range delays via thin-shell mapping and evaluated models using **Matplotlib**.
- Evaluated geomagnetic storm-event detection capabilities (POD, Critical Success Index, F1-score) alongside forecast errors, measuring system reliability in predicting solar disturbance impacts on positioning accuracy.

Codewithavra – Personal Portfolio Website [GitHub](#) [Live](#) Apr 2026 – May 2026

- Crafted a visually polished portfolio with **React 19** and **TypeScript**, implementing dark/light mode theming, fluid text animations, and scroll-driven transitions using **Motion** and **Lenis** smooth scroll.
- Engineered a fully responsive, performant UI with **Tailwind CSS v4**, focusing on micro-interactions, typography, and visual hierarchy to deliver a production-grade frontend experience.
- Deployed on **Vercel** with CI/CD via GitHub and integrated **Vercel Analytics** for real-time visitor insights.

ACHIEVEMENTS & CERTIFICATIONS

NPTEL – Programming in Java (IIT Kharagpur) [Certificate](#) Jul – Oct 2025

- Ranked in Top 1% (99%) among 26,183 learners; 12-week course covering OOP, exception handling, and Java application development.

Full Stack Development with MEAN Stack — Ardent Computech [Certificate](#) Jul 2025

- 36-hour training covering MongoDB, Express.js, Angular, and Node.js for full-stack application development.

NPTEL – Introduction to Database Systems (IIT Madras) [Certificate](#) Jan – Apr 2025

- Ranked in Top 1% (90%) among 2,768 participants; 12-week course covering relational models, SQL, normalization, and transaction management.

Finalist – CIRCUITHON 2024 (IEEE IEM CAS Student Branch) [Certificate](#) Dec 2024

- Selected as finalist among 150 participants in a circuit design competition.

EDUCATION

Narula Institute of Technology, Kolkata June 2023 – Present
B.Tech in Electronics & Communication Engineering *CGPA: 9.39*

Kalyan Nagar Vidyapith, Kolkata 2022
10+2 (W.B.C.H.S.E)

Kalyan Nagar Vidyapith, Kolkata 2020
10 (W.B.B.S.E)